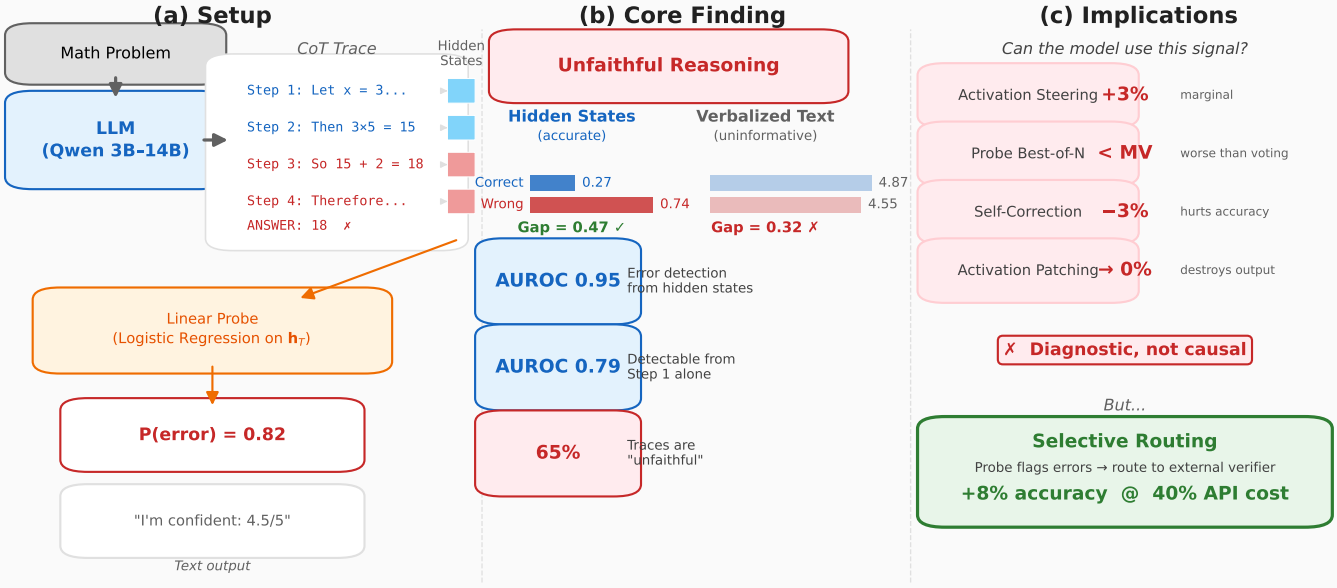

(a) Setup
Math Problem
LLM
(Qwen 3B–14B)
CoT Trace
Hidden States
Step 1: Let x = 3...
Step 2: Then 3×5 = 15
Step 3: So 15 + 2 = 18
Step 4: Therefore...
ANSWER: 18 ✗
Linear Probe
(Logistic Regression on $\mathbf{h}_T$)
P(error) = 0.82
"I'm confident: 4.5/5"
Text output
(b) Core Finding
Unfaithful Reasoning
Hidden States
(accurate)
Verbalized Text
(uninformative)
Correct 0.27
Wrong 0.74
Gap = 0.47 ✓
4.87
4.55
Gap = 0.32 ✗
AUROC 0.95
Error detection from hidden states
AUROC 0.79
Detectable from Step 1 alone
65%
Traces are "unfaithful"
(c) Implications
Can the model use this signal?
Activation Steering +3% marginal
Probe Best-of-N < MV worse than voting
Self-Correction −3% hurts accuracy
Activation Patching→ 0% destroys output
✗ Diagnostic, not causal
But...
Selective Routing
Probe flags errors → route to external verifier
+8% accuracy @ 40% API cost